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Chronic pain produces hypervigilance to predator odor in mice

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The adaptive significance of acute pain (to withdraw from tissue-damaging or potentially tissue-damaging external stimuli, and to enhance the salience of the stimulus resulting in escape and avoidance learning) and tonic pain (to enforce recuperation by punishing movement) are well-accepted [1]. Pain researchers, however, generally assert that chronic pain has no adaptive significance, representing instead a pathophysiological state. This belief was recently challenged by the observation [2] that nociceptive sensitization caused by a chronic pain-producing injury reduced predation risk in squid (*Doryteuthis pealeii*). In that study, injury to an arm (removal of the tip with a scalpel) 6 hours prior led to increased targeting by black sea bass, resulting in decreased survival of the squid in a 30-minute trial featuring free interaction between predator and prey. The surprising finding was that anesthesia during surgery, preventing the chronic nociceptor sensitization associated with such injuries, led to even lower probability of survival. That is, the likely presence of pain increased apparent fitness, and the authors concluded that the chronic pain state and its associated nociceptive sensitization represented an adaptive function. Pain-induced defensive behaviors affecting fitness have also been reported in crustaceans (*Gammarus fossarum*) [3]. It is, however, currently unknown whether this may also be true in any other species, including in Mammalia.

We designed an octagonal O-maze (Figure 1A and Supplemental Experimental Procedures) in which food-deprived mice were placed in one octant. A food reward was placed ~30 cm away via the 'short route'; alternatively, mice could walk all the way around the maze (the 'long route'; ~200 cm) to retrieve the reward. Mice were trained for 6–8 days until they opted for the short route in >80% of 10 consecutive rewarded trials. The maze was designed with a fan and vacuum system such that volatilized odors could be introduced into the maze, but were restricted to the octant corresponding to the short route. On the single testing day immediately following training we found that fox urine caused mice to avoid the short route in favor of the long route in a concentration-dependent manner ($n=5-11$; main effect of concentration: $F_{2,19}=7.4$, $p=0.004$; Figure 1B). This shift represents vigilance to potential predation, as the long route minimizes exposure to the fox odor. We avoided the use of yet higher volumes of fox urine to prevent spread of the odor beyond the short-route octant, and to prevent the possibility of floor effects.

Mice received sham surgery or spared nerve injury (SNI) 1–2 weeks prior to maze training and 2–3 weeks prior to maze testing by experimenters blinded

to surgical status. SNI is a preclinical model of complex regional pain syndrome, type 2 (causalgia), involving transection of two of the three distal branches of the sciatic nerve innervating the hind paw. SNI is associated with nociceptive sensitization and robust and long-lasting neuropathic pain-related behaviors [4,5]. A two-way ANOVA revealed a significant odor x surgery interaction ($n=11-13$; $F_{1,44}=6.6$, $p=0.01$). Mice given SNI displayed significantly more avoidance of the short route than those given sham ($p=0.037$), and thus were hypervigilant to the fox odor (Figure 1C).

Most pain research is aimed at providing proximate explanations (regarding mechanism and ontogeny) for pain — especially the more pressing clinical concern of chronic pain — leaving two of Tinbergen's four questions [6] regarding adaptive significance and phylogeny largely unaddressed. In fact, the leading hypothesis regarding the ultimate explanations for chronic pain is the 'smoke detector principle', which posits that excessive pain reporting and defensive responses to injury long past tissue healing are akin to a broken, or oversensitive, smoke detector, which continues to sound the alarm despite the lack of ongoing threat [7]. The observations of Crook

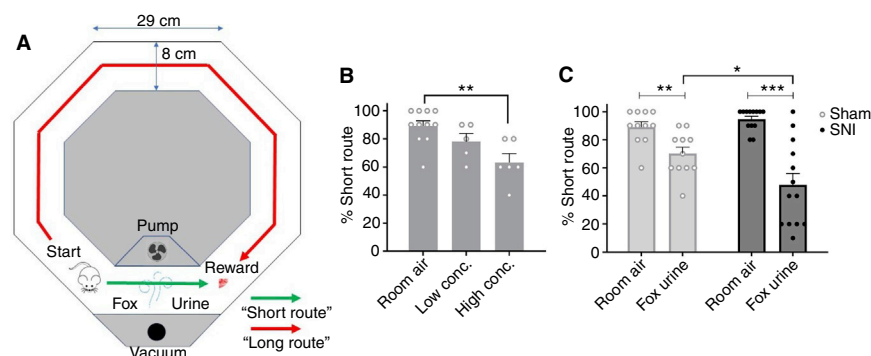


Figure 1. Hypervigilance to predator odor in mice produced by spared nerve injury (SNI). (A) Cartoon description of the octagonal O-maze used for training and testing. Mice were placed in the start octant and could choose to retrieve a reward via short or long routes. During testing, fox urine was pumped into and vacuumed out of the octant representing the short route; avoidance of this route would thus represent vigilance towards possible predation. (B) Concentration-dependent avoidance of the maze short route infused with fox urine. Bars represent mean $\pm$ SEM percentage of ten trials in which the mouse took the short route (that is, through the odor) to the reward. Trials in separate groups of mice featured no fox urine (room air pumped through; $n=11$), or fox urine at low concentration (Low Conc.; 1 ml of 100% fox urine over ~10 min; $n=5$) or at high concentration (High Conc.; 2 ml; $n=7$). ** $p<0.01$ by Tukey's post-hoc test. (C) Mice experiencing chronic pain from SNI display hypervigilance to fox urine. Bars as in graph (B) ($n=11-13$ mice/condition); mice received SNI or sham surgery 2–3 weeks prior to testing in a maze featuring room air or 2 ml fox urine. * $p<0.05$, ** $p<0.01$, *** $p<0.001$ by Student's t -tests as indicated.

and colleagues [2] in squid represent the first empirical data to suggest that chronic pain may indeed have adaptive significance. By continually reminding an injured organism of its increased risk of predation, chronic pain may increase vigilance, leading to increased Darwinian fitness. Ethical considerations precluded us from testing actual predation of mice, and thus fitness could not be evaluated. However, we believe that the hypervigilance to fox odor demonstrated here is a close, experimentally tractable proxy that mirrors in some respects the increased defensive behaviors displayed by injured squid [2]. We suggest, therefore, that the chronic-pain hypervigilance hypothesis of Crook *et al.* [2] may hold in mammals.

SUPPLEMENTAL INFORMATION

Supplemental Information includes the experimental procedures and supplemental reference, and can be found with this article online at <https://doi.org/10.1016/j.cub.2020.06.025>.

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Active escape of prey from predator vent via the digestive tract

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Predation pressures can lead to the evolution of escape behavior in prey animals [1,2]. Most previous studies investigated how prey can escape from predators before contact [1,2], whereas recent studies have focused on the post-contact escape of prey [2]. Predators can damage prey by biting or chewing, and the predator's digestive system ultimately kills almost all prey after swallowing. However, several species can survive passage through a predator's gut and are ultimately excreted with feces [3–5]. Such escapes from the predator's vent (cloaca or anus) are considered passive [3–5]. Survival in the extreme pH and anaerobic conditions of the predator digestive tract depends on the speed of passage and activity of the prey. This lethal environment may impose selective pressures on the evolution of quick and active escape behavior in swallowed prey species. Here, I report active escape of the aquatic beetle *Regimbartia attenuata* (Coleoptera: Hydrophilidae) from the vents of five frog species via the digestive tract. Although adult beetles were easily eaten by frogs, 90% of swallowed beetles were excreted within 6 h (0.1–6.0 h) after being eaten and, surprisingly, were still alive. When beetle legs were experimentally fixed with wax, all of the treated beetles were killed in the frogs' digestive system and finally excreted >24 h (38.3–150.3 h) after consumption. Therefore, swallowed beetles likely used their legs to move through the digestive tract toward the frog vent, hastening their escape. This study is the first to document active prey escape from the vent of a predator and to show that prey may promote predator defecation to hasten escape from inside the predator's body.

R. attenuata is common in wetlands, where the pond frog *Pelophylax*

nigromaculatus (Anura: Ranidae) is abundant (Figure 1A,B) [6,7]. This frog species is a potential predator of *R. attenuata*, because it eats both terrestrial and aquatic insects in the wetland environment [7,8]. To investigate the responses of *R. attenuata* to frog predation, I provided *R. attenuata* to *P. nigromaculatus* under laboratory conditions (25°C; see Supplemental Information). All *P. nigromaculatus* swallowed *R. attenuata* (n = 15). However, 93.3% of swallowed *R. attenuata* were excreted within 4 h (mean ± standard error: 1.6 ± 0.2 h) after being swallowed (Figure 1C,D; Video S1). The excreted beetles were frequently entangled in fecal pellets (Video S1). Beetles recovered immediately after being excreted, and thereafter swam actively. The same escape behavior was observed when *R. attenuata* were swallowed by four other frog species (Figure 1D); the escape success rates were 66.7% from *Pelophylax porosus* (n = 3; time from swallowing to excretion: 1.2 ± 0.3 h), 75.0% from *Glandirana rugosa* (n = 4; 2.2 ± 0.5 h), 87.5% from *Fejervarya kawamurai* (n = 8; 1.2 ± 0.2 h), and 100.0% from *Hyla japonica* (n = 10; 2.7 ± 0.6 h; Video S1). Of the excreted beetles, 94.4% survived for at least 2 weeks (15–187 days) following excretion. Killed *R. attenuata* were eventually excreted as feces >48 h (88.6 ± 25.5 h; n = 4) after being swallowed. When *P. nigromaculatus* was provided a different aquatic beetle, *Enochrus japonicus* (Coleoptera: Hydrophilidae), all swallowed beetles (n = 13) were killed and eventually excreted >24 h (49.9 ± 8.7 h) after consumption (Figure 1E).

Adult *R. attenuata* use their legs to swim actively [6]. To test whether the legs of *R. attenuata* play an important role in escaping a frog's gut, I provided *P. nigromaculatus* with adult *R. attenuata* whose legs had been experimentally fixed with sticky wax (n = 15). All of the treated beetles were killed and eventually excreted as feces >24 h (58.4 ± 7.0 h, n = 15) after being swallowed by *P. nigromaculatus* (Figure 1E). Control *R. attenuata* individuals that survived passage through the frog gut were excreted much more rapidly than treated beetles (1.6 ± 0.2 vs. 58.4 ± 7.0 h; Figure 1E; Table S1).